SEMI-MARKOV MODELS IN ECONOMY: ANALYZING EFFECTIVENESS IN ECONOMIC PROCESSES

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ABSTRACT

Semi-Markov models extend classical Markov chains by allowing general distributions of sojourn times, thus capturing the irregular timing of economic transitions. Unlike standard models with memoryless assumptions, semi-Markov processes incorporate duration dependence and heterogeneity across time. This makes them highly suitable for analyzing labor dynamics, business cycles, credit risk, and production reliability. By enhancing forecasts and reducing uncertainty, semi-Markov methods strengthen policy design, improve cost efficiency, and contribute to long-term economic resilience.

1. INTRODUCTION

Modern economies operate under uncertainty, where firms, households, and institutions experience state changes governed by stochastic and time-dependent processes. Classical Markov chain models have long been used to study these phenomena, yet their assumption of exponential, memoryless holding times rarely fits empirical reality.

Semi-Markov models (SMMs) overcome this by assigning flexible distributions to the duration spent in each state. This enables richer temporal dynamics, capturing persistence, path-dependence, and the variable timing that characterize real economies. Their applications now span from financial stability modeling and credit risk assessment to labor market transitions and macroeconomic forecasting.

2. METHODOLOGICAL FOUNDATIONS

A semi-Markov (SM) process $\{X(t), t \geq 0\}$ operates on a finite or countable state space $S = \{1, 2, \ldots, n, \ldots\}$. Transition probabilities p_{ij} determine the likelihood of moving from state i to j , while sojourn times follow general probability distributions $F_{ij}(t)$.

This dual specification captures both which transition occurs and when it occurs, allowing for non-exponential distributions such as Weibull, log-normal, or Pareto. Consequently, SMMs reproduce empirical waiting-time patterns more accurately than memoryless Markov chains. Although computationally more intensive, they yield results that better reflect real-world persistence and variability.

It should be mentioned, that transient investigations of SMM are very difficult. There exist several methods for such investigations, but they are all solutions to complicated equations and are not available in a convenient form.

Our approach is building upon a completely novel, simple, and very effective, partially probabilistic method for investigating semi-Markov models earlier developed by team of Georgian scientists. This method represents a special variety of the supplementary variables technique (SVT) — a modified SVT avoiding partial differential equations (Kolmogorov equations). This greatly enhances simplicity and effectiveness. Leading experts note that it will significantly simplify the investigation of both SMMs constructed in our research and SM random processes in general. We will demonstrate the capabilities of this method for all our SM models (as well as for several models of interest constructed and investigated by other leading researchers. We aim to demonstrate the simplicity and effectiveness of the technique.

3. KEY APPLICATIONS IN ECONOMIC PROCESSES

Labor Market Dynamics: SMMs model transitions between employment, unemployment, and inactivity, accounting for the fact that joblessness varies in duration. This enables improved evaluation of active labor market policies and retraining programs.

Business Cycles and Macroeconomic Stability: Traditional models treat expansions and recessions as memoryless processes, but economic phases display duration dependence. SMMs allow sojourn times in each phase to follow flexible distributions, providing more realistic estimates of turning points and persistence.

Financial Risk Assessment: In credit risk modeling, borrower default probability depends on both credit quality and time spent in a given rating. SMMs incorporate this time-dependence by fitting distinct sojourn-time distributions for different credit grades, yielding more accurate default and loss estimates.

Production Systems and Supply Chains: SMMs quantify machine breakdowns, recovery times, and inventory transitions, improving maintenance planning and system reliability.

4. COMPARATIVE EFFECTIVENESS

The effectiveness of semi-Markov models stems from several characteristics:

- Temporal realism: Incorporation of empirically observed distributions of waiting times.
- Improved forecasting: More precise estimation of business-cycle peaks, credit defaults, and labor turnover.
- Policy evaluation: Enables quantitative assessment of program durations and welfare dependencies.
- Resilience analysis: Measures system reliability under uncertainty, supporting investment and risk-mitigation decisions.

Compared with classical Markov chains, SMMs require more data and computational effort but produce insights aligned with observed behavior.

5. BUSINESS CYCLES AND FINANCIAL RISK: INTEGRATED PERSPECTIVE

For macroeconomic analysis, SMMs capture the growing fragility of expansions and the deepening persistence of recessions over time. By estimating hazard functions for transitions between "expansion" and "recession" states, analysts can anticipate reversals and calibrate counter-cyclical policies.

In finance, similar duration dependence explains why default probabilities increase with prolonged exposure to low-grade credit states. SMM-based rating transition matrices, calibrated using empirical data, significantly enhance Value-at-Risk (VaR) and Expected Loss (EL) metrics, as well as pricing of credit derivatives. Applications in emerging economies like Georgia show additional relevance: such economies face irregular cycles due to external shocks, and SMMs help capture this variability more effectively than traditional models.

6. ECONOMIC IMPLICATIONS: RISK REDUCTION AND EFFICIENCY GAINS

Risk Reduction: Earlier recognition of turning points reduces the depth and duration of recessions. More accurate estimation of default likelihood prevents systemic under-assessment of risk. Modeling unemployment durations refines targeting of assistance programs, and anticipation of downtime lowers operational volatility.

Cost Reduction and Efficiency: Improved forecasting reduces unnecessary expenditure and enhances allocative efficiency. Banks lower capital buffer costs through better risk modeling. Governments optimize welfare budgets via accurate unemployment duration estimates. Firms reduce maintenance and inventory costs through probabilistic optimization. Altogether, SMMs facilitate resource allocation that minimizes waste and maximizes welfare, key to achieving both allocative and dynamic efficiency.

7. CONCLUSION

Semi-Markov models bridge the gap between simplified Markovian frameworks and the complexity of real economies. By incorporating flexible time distributions, they deliver more accurate forecasts of transitions in labor, financial, and production systems. Their integration into macroeconomic and financial analysis strengthens resilience, reduces costs, and enhances long-term economic efficiency.

For economies such as Georgia, exposed to external shocks and structural vulnerabilities, semi-Markov modeling offers a methodological advantage for designing adaptive, data-driven policies that sustain growth and stability.

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